

Organism Discovery Report

This Organism Discovery Report (ODR) was generated by TaxTriage, a flexible, containerized bioinformatics pipeline for pathogen detection from untargeted DNA or RNA sequencing data. TaxTriage supports both short-read (Illumina) and long-read (ONT, PacBio) sequencing platforms and incorporates quality control, organism classification, read mapping, and a unified confidence metric for all detected organisms.

The ODR provides a consolidated summary of the evidence supporting pathogen identification, including key results from the intermediate analyses performed throughout the pipeline. TaxTriage is intended for broad deployment and early-stage outbreak investigations and is not intended as a standalone diagnostic capability.

Summary Statistics

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Number of Samples

158,510

Total Reads Analyzed

52.7%

% of Reads Aligned

23

Unique Organisms

26

Unique Genera

8

High Consequence
Pathogens Detected



Color Key

	Category	Description
	Primary Pathogen (Matched Sample Type)	Detected by taxonomic id or assembly name and classified as clinically relevant for this sample type.
	Primary Pathogen (Other Sample Type)	Detected and classified as clinically relevant for a different sample type.
	Species/Subkey Summary (inferred)	Species-level detection exceeds the threshold, but individual strains do not. (!) indicates the highest scoring primary strain below the threshold.
	Potential	Potential pathogen requiring further investigation
	DNA Pathogen	Detected pathogen has a DNA genome.
	RNA Pathogen	Detected pathogen has an RNA genome.
	High Consequence	Organism of elevated clinical significance that may warrant immediate review.
	[site flora]	Identifies organisms that are normal flora at the indicated body site but were detected in a normally sterile sample (blood, CSF, or serum).

Categories excluded by report settings and omitted from this report: **Commensal**, **Opportunistic**, **Unknown**. See the output reports or multi-run analysis for additional details.

Column Explanations

Organism	Detected organism with associated annotation, taxonomic ID, and taxonomic rank.
TASS Score	Confidence score for organism detection (0–100), with higher values indicating greater confidence.
Classifier Reads	Number of reads assigned to the organism by Kraken2/Centrifuge.
Aligned Reads	Number of reads aligned to the organism's reference genome. (%) indicates the proportion of total sample reads aligned to that organism. Aligned read counts may differ from the original mapping results due to post-processing steps that remove potential false-positive alignments.
RPM	Reads Per Million (RPM), a normalized abundance metric that enables comparison across samples.
% Coverage	Percentage of the organism's genome covered by aligned reads.
High ANI Matches	Lists strains with Average Nucleotide Identity (ANI) above the specified threshold.
Control Comparison	Displays an organism's TASS score relative to control samples. • Red: Negative-control range. • Green: Positive-control range. • Purple: In-silico simulation range, when available. • x: Sample TASS fold change; x rd: Read-count fold change. • Prefix symbols: – max negative control, + min positive control, ∞ in-silico control.

Run Summary

Sample	Sample Type	Platform	Total Reads	Aligned Reads	Organism Reads
Miseq_Run_A	nasal	illumina	117,338	63,364	63,364
Nanopore_Run_A	stool	oxford	3,000	1,991	1,991
Negative_Control_Miseq	nasal	illumina	172	109	109
Positive_Control_Miseq	nasal	illumina	38,000	18,021	18,021

Per-Sample Classification Results

Sample 1: Miseq_Run_A							
SAMPLE TYPE		SEQUENCING PLATFORM	TOTAL READS	ALIGNED READS	SPECIES GROUPS	TASS CUTOFF	
nasal		illumina	117,338	63,364	9	75.0	

	Organism	TASS Score	Classifier Reads	Aligned Reads	RPM	% Coverage	Control Comparison
	Orthopneumovirus 1 species; 2 qualifying strains						
	Orthopneumovirus hominis ■ (3049954) species	100.0	70	237 (0.2%)	2,609	100.0%	+ 1.0×4.7× rd
	! Human respiratory syncytial virus A ■ (208893) qualifying strain	100.0	48	193 (0.2%)	3,046	100.0%	+ 1.0×3.9× rd
	! human respiratory syncytial virus (B) ■ (11250) qualifying strain	92.4	70	44 (0.0%)	694	58.7%	+ 0.9×0.9× rd
	Betacoronavirus 1 species; 1 qualifying strain						
	Severe acute respiratory syndrome coronavirus 2 ★ ■ (2697049) strain	100.0	46	92 (0.1%)	1,452	65.3%	-
	Listeria 1 species; 2 qualifying strains						
	Listeria monocytogenes EGD-e ★ ■ (169963) strain	99.8	4,731	9,243 (7.9%)	145,871	61.1%	+ 2.4×99.4× rd - 4.4×1155.4× rd
	Orthopoxvirus 1 species; 1 qualifying strain						
	Monkeypox virus ★ ■ (10244) strain	98.6	635	1,191 (1.0%)	18,796	79.7%	+ 1.2×148.9× rd
	Neisseria 1 species; 1 qualifying strain						
	Neisseria gonorrhoeae ■ (485) species	92.9	3,486	6,972 (5.9%)	104,065	61.9%	+ 1.0×1.0× rd - 4.5×871.5× rd
	Escherichia 1 species; 1 qualifying strain						
	Escherichia coli O157:H7 ★ ■ (83334) strain	92.6	8,939	16,740 (14.3%)	264,156	54.1%	+ 1.3×7.9× rd - 5.2×930.0× rd
	Pseudomonas 1 species; 2 qualifying strains						
	Pseudomonas aeruginosa PA1 ■ (1279007) strain	80.8	20,125	7,834 (6.7%)	123,635	22.2%	+ 4.3×191.1× rd - 11.3×3917.0× rd
	Positive Control Organisms Not Detected in Sample: Enterococcus (1350) Control TASS 40.85, 81 reads, source: Positive_Control_Miseq.paths.json; Limosilactobacillus (2742598) Control TASS 89.97, 70 reads, source: Positive_Control_Miseq.paths.json; Salmonella (590) Control TASS 16.39, 191 reads, source: Positive_Control_Miseq.paths.json; Saccharomyces (4930) Control TASS 98.05, 386 reads, source: Positive_Control_Miseq.paths.json						

Sample 2: Nanopore_Run_A						
SAMPLE TYPE	SEQUENCING PLATFORM	TOTAL READS	ALIGNED READS	SPECIES GROUPS	TASS CUTOFF	
stool	oxford	3,000	1,991	9	75.0	
	Organism	TASS Score	Classifier Reads	Aligned Reads	RPM	% Coverage
	Listeria 1 species; 1 qualifying strain					
	Listeria monocytogenes EGD-e ★ ■ (169963) strain	99.8	199	199 (6.6%)	99,950	29.6%
	Salmonella 1 species; 3 qualifying strains					
	Salmonella enterica subsp. enterica serovar Typhimurium str. LT2 ★ ■ (99287) strain	77.8	42	39 (1.3%)	10,573	3.7%
	Escherichia 1 species; 1 qualifying strain					
	Escherichia coli ★ ■ (562) species ! Escherichia coli O157:H7 TASS 69.7 19 reads	89.2	136	251 (8.4%)	46,598	15.1%
	Pseudomonas 1 species; 2 qualifying strains					
	Pseudomonas aeruginosa ■ (287) species ! Pseudomonas aeruginosa PA1 TASS 44.3 3 reads	87.2	8	17 (0.6%)	1,306	1.7%

Sample 3: Negative_Control_Miseq

SAMPLE TYPE	SEQUENCING PLATFORM	TOTAL READS	ALIGNED READS	SPECIES GROUPS	TASS CUTOFF
nasal	illumina	172	109	8	75.0

No organisms met the reporting confidence threshold for this sample.

6 high-consequence detections were identified below the threshold. See the Low Confidence, High Consequence Detections section for details.

Sample 4: Positive_Control_Miseq						
SAMPLE TYPE	SEQUENCING PLATFORM	TOTAL READS	ALIGNED READS	SPECIES GROUPS	TASS CUTOFF	
nasal	illumina	38,000	18,021	12	75.0	
	Organism	TASS Score	Classifier Reads	Aligned Reads	RPM	% Coverage
	Orthopneumovirus 1 species; 1 qualifying strain					

	Organism	TASS Score	Classifier Reads	Aligned Reads	RPM	% Coverage
	human respiratory syncytial virus (B) ■ (11250) strain	99.0	25	50 (0.1%)	2,775	56.7%
	Neisseria 1 species; 1 qualifying strain					
	Neisseria gonorrhoeae ■ (485) species	92.9	3,493	6,986 (18.4%)	368,847	61.5%
	Orthopoxvirus 1 species; 1 qualifying strain					
	Monkeypox virus ★ ■ (10244) strain	80.4	4	8 (0.0%)	444	1.2%
	Escherichia 1 species; 1 qualifying strain					
	Escherichia coli ★ ■ (562) species ! Escherichia coli O157:H7 TASS 73.1 2,130 reads	96.2	7,541	14,380 (37.8%)	388,518	43.2%
	Saccharomyces 1 species; 1 qualifying strain					
	Saccharomyces cerevisiae ■ (4932) species ! Saccharomyces cerevisiae S288C TASS 51.8 368 reads	98.0	193	386 (1.0%)	1,551	1.3%

Low Confidence, High Consequence Detections

Detected strains below the confidence threshold of 75.0.

	Sample	Organism	TASS Score	Classifier Reads	Aligned Reads
	Positive_Control_Mi seq	Listeria monocytogenes EGD-e (169963) ★	42.1	49	93
	Positive_Control_Mi seq	Salmonella enterica (28901) ★	9.6	79	2
	Positive_Control_Mi seq	Salmonella enterica subsp. enterica serovar Typhimurium (90371) ★	6.2	79	1
	Positive_Control_Mi seq	Salmonella enterica subsp. enterica serovar Typhimurium str. LT2 (99287) ★	5.4	79	1
	Negative_Control_M iseq	Listeria monocytogenes EGD-e (169963) ★	22.5	4	8
	Negative_Control_M iseq	Escherichia coli O157:H7 (83334) ★	17.9	9	18

Additional Information

- Sterile samples (**blood, CSF, sterile, and serum**) may yield lower TASS scores due to relatively low read count or limited genomic coverage. All detected pathogens for these sample types are therefore classified as **primary pathogens** by default.
- Sample names and species groups throughout the report are hyperlinked for navigation. Selecting a sample or organism entry will jump directly to its corresponding section in the document.
- Organism groups are sorted by TASS score by default, or alphabetically when the alphabetical sorting option is enabled.
- Only samples and organism groups containing visible qualifying strains are included in the index and navigation sections.
- When subkey grouping is enabled, each genus-level group expands into species/subkey summary rows followed by qualifying child strains that also pass the reporting threshold.
- Low-confidence, high-consequence detections that do not appear in the PDF report may still be present in the corresponding Discovery Analysis TXT output file.
- Please visit our [DOCUMENTATION PAGE](#) for additional details on TASS confidence scoring.

Questions, feedback, or issues related to this report or the TaxTriage workflow can be submitted through the GitHub repository. General questions and discussions should be posted in the [Discussions](#) section, while bugs and feature requests should be submitted through the [Issues](#) tracker.

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Samples in Run:

Format: Sample name (# Total Alignments;# Primary Pathogens) → Category Label ■ (# Primary Strains, Max TASS)

[Miseq_Run_A](#) — illumina (111,674.0 alignments;10 primary pathogens)

→ Orthopneumovirus ■ (2, 100.0), Betacoronavirus ■ (1, 100.0), Listeria ■ (2, 99.8), Orthopoxvirus ■ (1, 98.9), ... (3 more)

[Nanopore_Run_A](#) — oxford (5,690.0 alignments;7 primary pathogens)

→ Listeria ■ (1, 99.8), Salmonella ■ (3, 95.3), Escherichia ■ (1, 91.2), Pseudomonas ■ (2, 88.4)

[Positive_Control_Miseq](#) — illumina (36,070.0 alignments;11 primary pathogens)

→ Orthopneumovirus ■ (1, 99.0), Neisseria ■ (1, 92.9), Orthopoxvirus ■ (1, 80.4), Escherichia ■ (1, 96.2), ... (1 more)

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